

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 365 trials
Total training paths	1 · 365 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	flatten → feature dim 15840
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=292 · test=73

Augmentation (training only)

Enabled	no
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (RBF)	0.9966	0.0103	1.0000
SVM (Linear)	1.0000	0.0000	1.0000
Random Forest	1.0000	0.0000	1.0000
KNN	1.0000	0.0000	1.0000
Logistic Regression	0.9966	0.0103	1.0000
LDA	1.0000	0.0000	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000
Stacking (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000

Per-class hold-out F1

Class	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting (soft)	Stacking	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	13
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	12
macro avg	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	73

External test (NewTestData)

Algorithm	External test acc
Random Forest	1.0000
LDA	1.0000
SVM (Linear)	0.8065
Logistic Regression	0.8065
Voting Soft (SVM+RF+KNN+LogReg)	0.6774
Stacking (SVM+RF+KNN+LogReg)	0.6774
SVM (RBF)	0.5806
KNN	0.4839

Per-class external accuracy

[illegible]